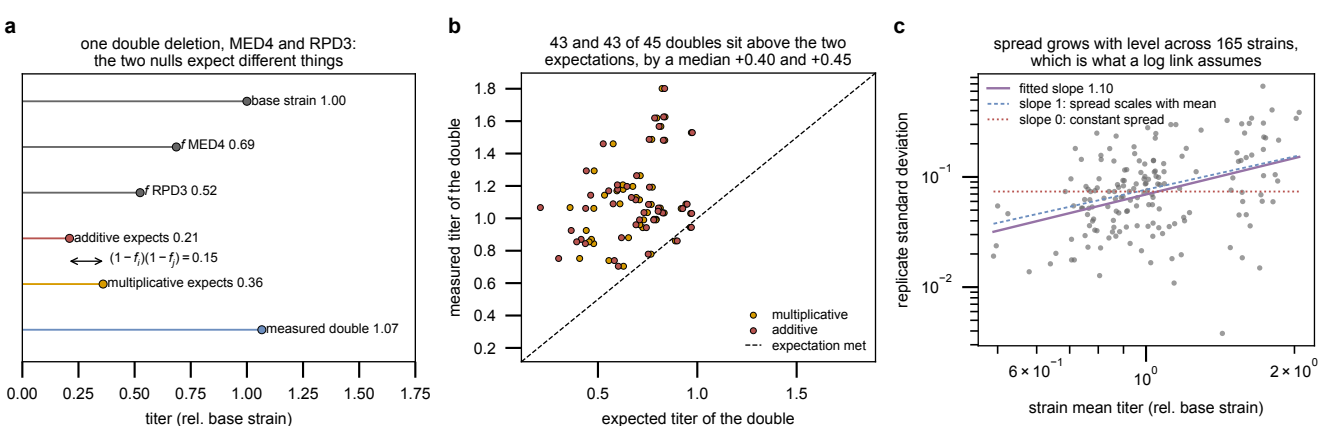
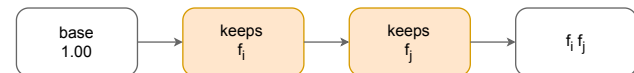


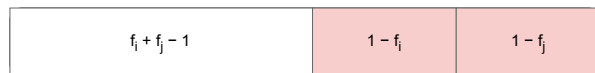


d effects are FRACTIONS: two steps in series



a fraction of a fraction: the second step only ever acts on what the first let through

effects are AMOUNTS: two draws on one pool



each loss is taken from the whole pool, so the part the first deletion already removed is counted a second time

multiplicative expects $(f_i + f_j - 1) + (1 - f_i)(1 - f_j)$: the two nulls differ by exactly the double-counted loss, which is why they disagree most when both single deletions are severe

e				
model	expects, for a double	residual measured on	fit to, and what that assumes	the reading that makes it the natural null
multiplicative	$f_i f_j$	linear titer	strain means, one standard error per strain, propagated by the delta method	each deletion keeps a fixed FRACTION of what reaches it, so independent steps of one flux compose; this is the null of the growth screens
additive	$f_i + f_j - 1$	linear titer	strain means, one standard error per strain, propagated by the delta method	each deletion removes a fixed AMOUNT from a shared pool, which is closer to how a titer in mg/L is built up
GLM log-link	$\exp(\alpha_i + \alpha_j)$, the multiplicative null	log titer	replicate titers, Gamma family: spread grows in proportion to the mean (panel c)	the readout is positive and noisier where it is larger, and every strain's replicates are used rather than its mean
log-OLS	$\exp(\alpha_i + \alpha_j)$, the multiplicative null	log titer	replicate log ratios to the base strain, constant spread on the log scale	the question is whether a deletion's FOLD effect carries into a new background, which is a log-scale question